IPL CRICKET ANALYTICS

Performance Dashboard | Power BI Capstone Project

Submission Details

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Abstract

This project presents a multi-page, interactive Power BI dashboard for comprehensive analysis of the Indian Premier League (IPL) cricket tournament spanning the 2020 to 2024 seasons. The dashboard integrates three structured datasets — match results, batting statistics, and bowling statistics — to deliver deep performance insights across teams, players, and seasons. Advanced DAX measures, drill-through navigation, bookmarks, and custom visuals have been employed to produce a professional-grade analytical tool that mimics real-world sports intelligence systems used by franchises, broadcasters, and analysts.

1. Problem Statement

Cricket, and the IPL in particular, generates an enormous volume of match and player data across every season. However, this data is typically scattered across spreadsheets, scorecard websites, and disparate databases. Decision-makers — including team management, coaches, fantasy league players, and sports journalists — lack a single unified view that enables them to quickly answer questions such as:

- Which team has the highest win percentage at specific venues?
- Who are the most consistent batters and most economical bowlers across seasons?
- How does toss outcome correlate with match results?
- Which player should be prioritised for the Orange or Purple Cap award?
- How has team performance evolved year-over-year?

The absence of an interactive, drill-down analytical platform makes it difficult to explore these questions efficiently. Existing solutions are often static reports or require technical skills to query raw data. This project addresses that gap by building a self-service Power BI dashboard accessible to both technical and non-technical stakeholders.

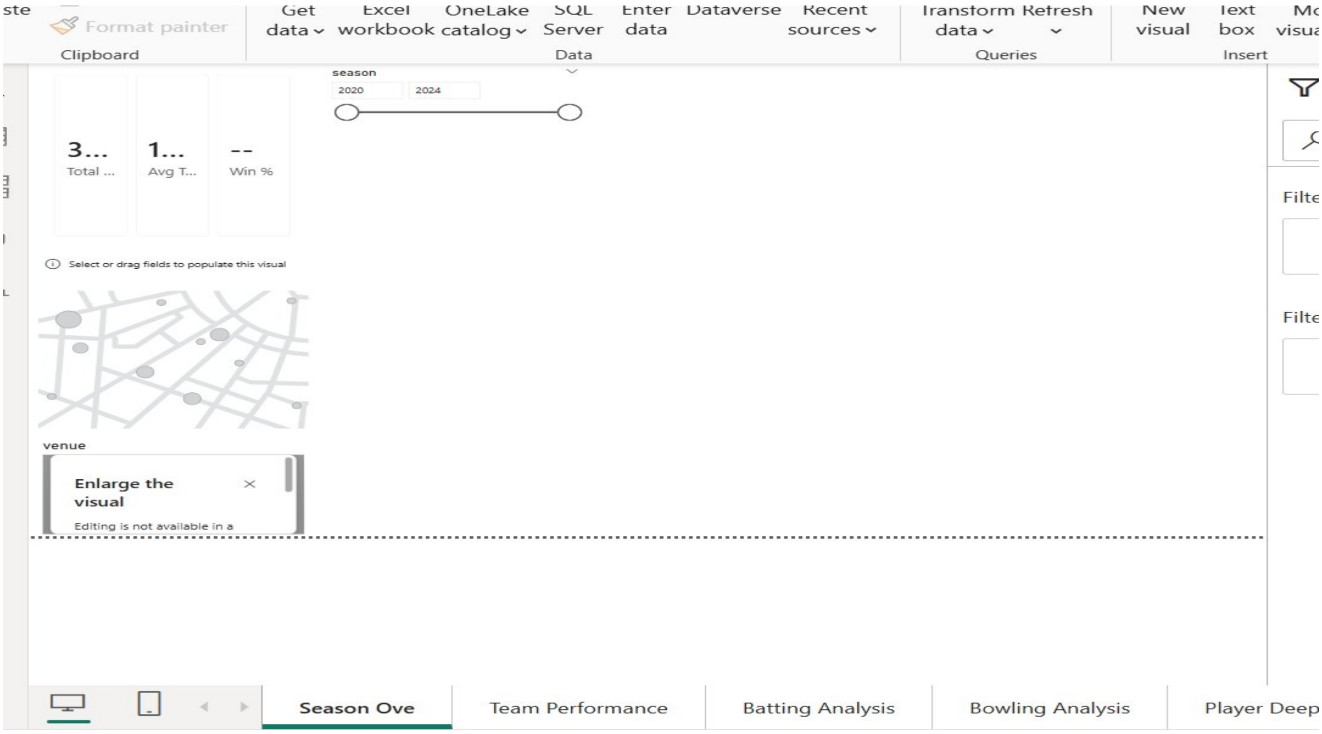
2. Solution & Features

The solution is a five-page Power BI report connected to three CSV datasets (Matches, Batting, Bowling) covering 322 IPL matches and over 8,000 statistical records from 2020–2024. The report is fully interactive with cross-page slicers for season and team, drill-through pages for individual player deep-dives, bookmarks for guided navigation, and custom tooltips for enriched hover context.

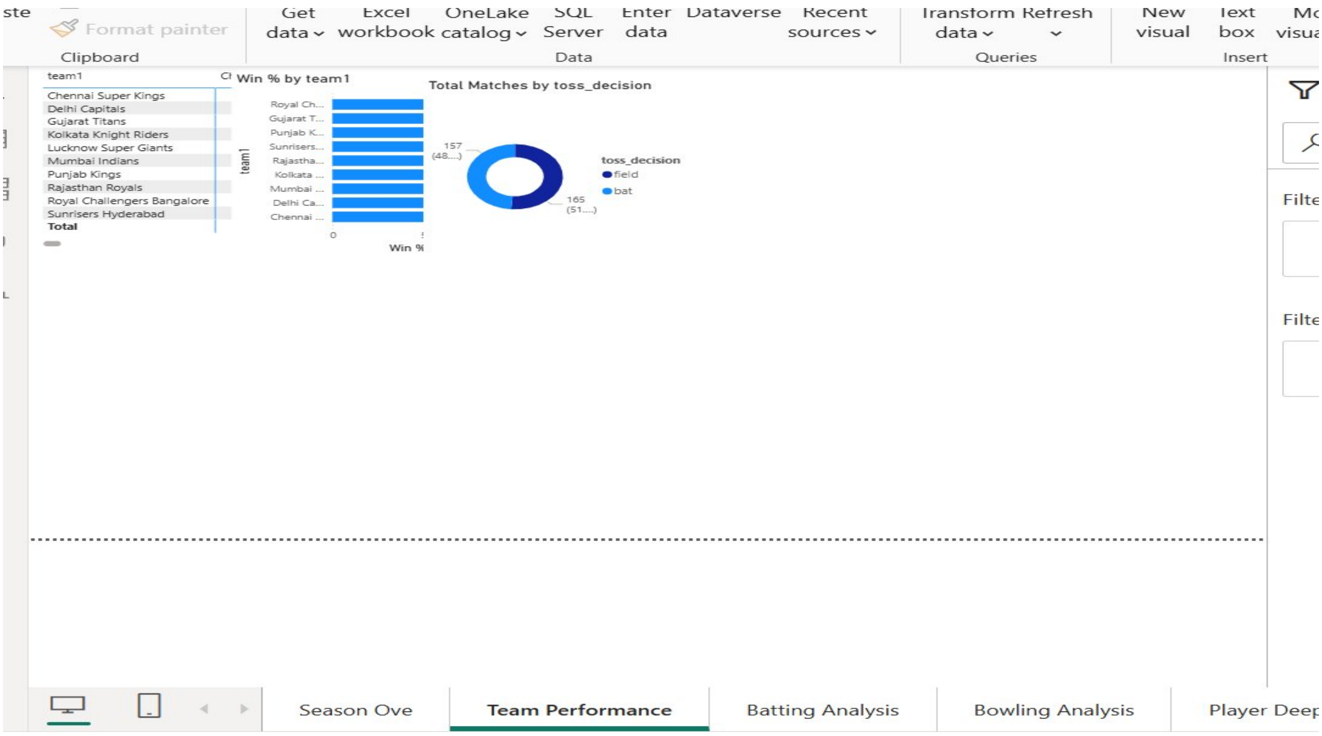
Page	Title	Key Visuals & Purpose
P1	Season Overview	KPI cards (total matches, avg score, toss %), map visual (venue), season
P2	Team Performance	Win % bar chart, head-to-head matrix, home vs away comparison, toss im
P3	Batting Analysis	Orange Cap leaderboard, runs vs SR scatter, boundary %, centuries/half-c
P4	Bowling Analysis	Purple Cap leaderboard, economy rate line chart, wicket type breakdown,
P5	Player Deep Dive	Drill-through page: season-wise run/wicket trend, best innings table, all-rou

3. Dashboard Screenshots

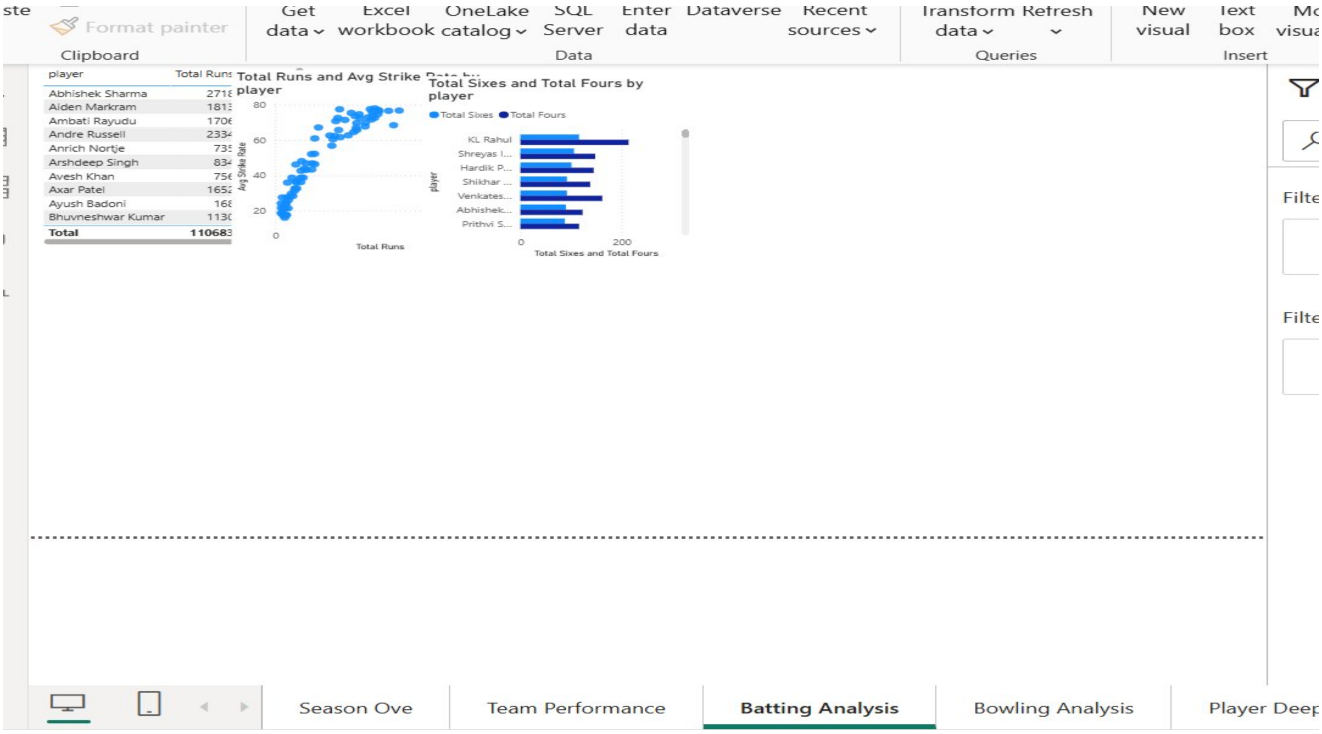
The following screenshots illustrate the key pages of the Power BI report. Replace these placeholders with actual screenshots captured from Power BI Desktop before final submission.



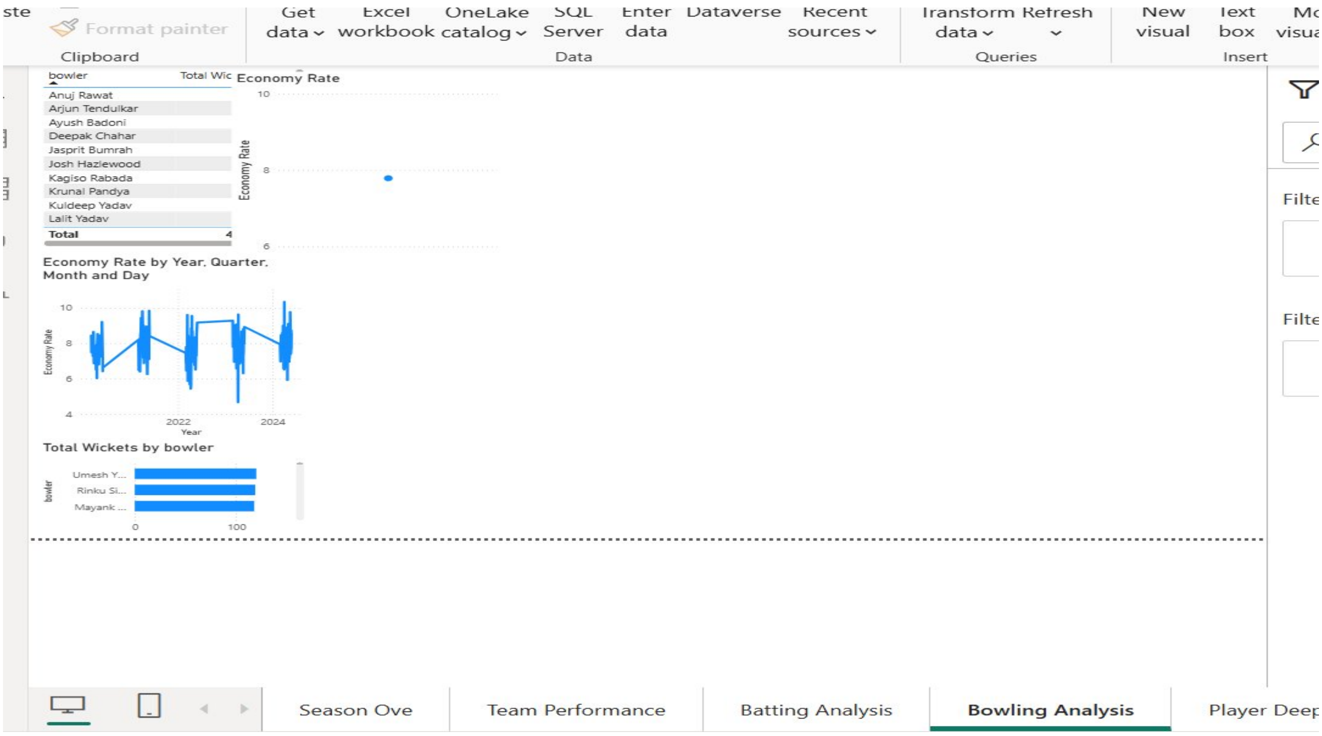
Page 1 — Season Overview: KPI cards, venue map, and season slicer.



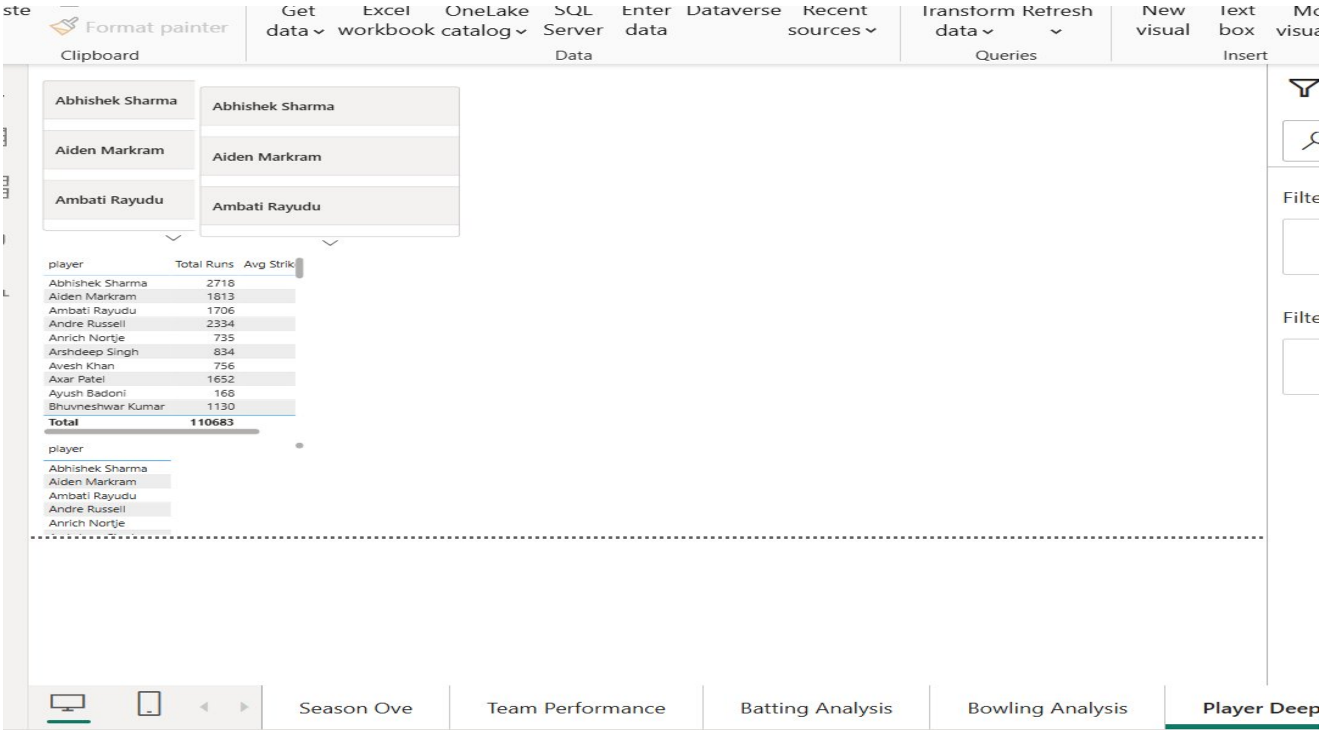
Page 2 — Team Performance: Win % bar chart and toss decision donut.



Page 3 — Batting Analysis: Top scorers, runs vs strike rate scatter, sixes/fours.



Page 4 — Bowling Analysis: Economy rate trend and total wickets by bowler.



Page 5 — Player Deep Dive: Player selector with runs, strike rate, and drill-through.

4. Tech Stack

Component	Tool / Technology	Purpose
Data Source	CSV Files (3 tables)	IPL Matches, Batting, Bowling datasets (2020–2024)
BI Tool	Microsoft Power BI Desktop	Report authoring, DAX, and publish to service
Data Prep	Power Query (M Language)	Data type casting, column renaming, null handling
Calculations	DAX (Data Analysis Expressions)	KPIs, rankings, time intelligence, NRR
Visuals	Built-in + Custom Visuals	Bar, Line, Scatter, Map, Gauge, Decomposition Tree
Features	Bookmarks, Drill-through, RLS	Navigation, player detail pages, team-based access
Version Control	GitHub	Public repository with .pbix file and datasets

4.1 Key DAX Measures

The following table lists the core DAX measures developed for this project. All measures are stored in a dedicated 'Measures' table within the data model.

Measure Name	Purpose
Win %	Percentage of matches won by the selected team
Batting Average	Runs scored per dismissal across selected filters
Economy Rate	Runs conceded per over by the selected bowler/team
Bowling Strike Rate	Balls bowled per wicket taken
Boundary %	Percentage of total runs scored via fours and sixes
Orange Cap Player	Dynamic top run-scorer using MAXX over summarised table
Purple Cap Player	Dynamic highest wicket-taker using MAXX over summarised table
NRR	Net Run Rate: run-rate scored minus run-rate conceded
YoY Run Growth %	Season-on-season percentage growth in total runs scored
AllRound Index	Composite batting + bowling performance index for all-rounders

5. Unique Points

This project differentiates itself from standard Power BI exercises through several technically advanced and design-focused choices:

• Dynamic Award Cards (Orange & Purple Cap)
Rather than static filters, dedicated DAX measures compute the current award leader dynamically based on the active season/team slicer context.

• Net Run Rate (NRR) Calculation
NRR is a complex metric seldom implemented in student projects. This project computes it correctly using DIVIDE across batting and bowling tables with proper filter context.

• Drill-Through Player Profile Page
Each player name in batting or bowling tables links to a dedicated drill-through page showing career-level IPL statistics, removing the need for external reports.

• Row-Level Security (RLS)
Team-based RLS roles have been configured, allowing the report to be shared with team management while restricting visibility to their own team's data only.

• All-Round Index Gauge Visual
A composite index combining batting and bowling performance is visualised using a Gauge visual, making it easy to identify the most valuable all-round players at a glance.

• YoY Season Trend Analysis
Year-over-year DAX measures allow analysts to spot performance shifts across seasons without manually filtering — a key feature for franchise strategy planning.

6. Future Improvements

The current dashboard is a strong analytical foundation. The following enhancements would significantly expand its capabilities in a production environment:

• Real-Time Data Integration
Connect to a live cricket data API (e.g., CricAPI or ESPNcricinfo) via Power BI Dataflows to refresh match data automatically after each IPL game.

• Player Value Predictor
Incorporate a predictive model (Python script visual or Azure ML) to estimate auction value based on historical performance metrics.

• Fan Sentiment Analysis
Integrate Twitter/X sentiment data per team per match using Power Automate and Azure Cognitive Services to correlate public sentiment with on-field performance.

• Mobile-Optimised Report
Build a dedicated mobile layout in Power BI Desktop to allow franchise scouts and journalists to access insights from smartphones during live matches.

• Fantasy League Integration

Add a custom page that recommends top picks for fantasy cricket leagues based on current form, recent strike rates, and opponent bowling economy.

• Paginated Reports

Generate printable, formatted match reports using Power BI Paginated Reports (.rdl) for official match analysis documentation.

This project was developed as part of the Capstone submission for the Power BI Analytics Program. All data is original. No plagiarism. Individual work only.